

COURSE CONTENT

Unit I: Random Signal Theory

Geometric representation of signals, Gram-Schmidt orthogonalization, Introduction to random variables, joint probability density function, characteristic functions, derived distributions, Introduction to random processes, Gaussian process, Linear functional of random process, Stationary and wide sense stationary random process, Power spectral density, White Gaussian noise.

Unit II: Base Band Transmission

Pulse code modulation (PCM), PCM Generation and Reconstruction, Quantization Noise, Non-Uniform Quantization and Commanding, Differential and adaptive PCM, Noise considerations in PCM, Delta modulation and demodulation, slope overload, Adaptive and Sigma Delta Modulation.

Unit III: Band Pass Transmission

Generation and detection of coherent & non-coherent Amplitude, Frequency & Phase Shift Keying (ASK, FSK, PSK), Quadrature modulation techniques, Minimum Shift Keying (MSK), M-ary FSK, Optimal receiver and Probability of error, comparison of various digital modulation techniques, Coherent reception of ASK, PSK and FSK, Non-Coherent reception of ASK, FSK, PSK and QPSK.

Unit IV: Information Theory and Source Coding

Definition of Information, Entropy, Mutual Information, Properties of Mutual Information, Relationship between entropy and mutual information, Types of Channels, Binary Symmetric Channel, Channel Capacity, Channel Coding Theorem, Shannon-Hartley Theorem, Kraft inequality, Huffman codes, Shannon-Fano coding.

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Unit V: Channel Coding

Linear Block Codes, Systematic linear codes and optimum decoding for the binary symmetric channel, Generator and Parity Check matrices, Syndrome decoding on symmetric channels; Hamming codes, Weight enumerators and the Mac-Williams identities; Perfect codes. Cyclic Codes, BCH codes, Convolution codes, Turbo codes.

COURSE CONTENT

Unit I: Introduction to Soft Skills and Professional Ethics

Aspects of Soft Skills: Effective Communication Skills, Personality Development, Importance of Professional Ethics.

Unit II: Team Building

Understanding nature of team: Mapping personal and professional goals of team members, Working effectively in a team through building relations and interpersonal communication. What is negotiation, Ways of negotiating, Understanding the power of language and non-verbal communication.

Unit III: Meetings and Presentation

How to call a meeting, how to organize a meeting, how to design the agenda and prepare minutes of the meeting. Researching for a presentation, structure of presentation, verbs often required, language focus, importance of body language in presentation, preparing an outline of a presentation, ending the presentation.

Unit IV: Stress Management and Time Management

Kinds of stress, Identifying the right reasons of stress, how to handle the pressure, Techniques to cope with the stressful situation at a workplace, Goal setting, Understand the importance of time and how to prepare the time line.

Unit V: Group Discussion and Public Speaking

Nature of discussion, Ways to form and present the arguments, public speaking skills and being successful in it.

Unit I: Human Values

Morals, values and Ethics, Integrity, Work ethic, Service learning, Civic virtue, Respect for others, Living peacefully, Caring, Sharing, Honesty, Courage, Valuing time, Cooperation, Commitment, Empathy, Self-confidence, Character, Spirituality: Introduction to Yoga and meditation for professional excellence and stress management.

Unit II: Engineering Ethics

Senses of 'Engineering Ethics', Variety of moral issues, Types of inquiry, Moral dilemmas, Moral Autonomy, Kohlberg's theory, Gilligan's theory, Consensus and Controversy, Models of professional roles, Theories about right action, Self-interest, Customs and Religion, Uses of Ethical Theories

Unit III: Engineering As Social Experimentation

Engineering as Experimentation, Engineers as responsible Experimenters, Codes of Ethics, A Balanced Outlook on Law.

Unit IV: Safety, Responsibilities And Rights

Safety and Risk, Assessment of Safety and Risk, Risk Benefit Analysis and Reducing Risk, Respect for Authority, Collective Bargaining, Confidentiality, Conflicts of Interest, Occupational Crime, Professional Rights, Employee Rights, Intellectual Property Rights (IPR) – Importance of Plagiarism, Discrimination

Unit V: Global Issues

Multinational Corporations, Environmental Ethics, Computer Ethics, Weapons Development, Engineers as Managers, Consulting Engineers, Engineers as Expert Witnesses and Advisors, Moral Leadership, Code of Conduct, Corporate Social Responsibility

COURSE CONTENTS

Unit I: Introduction to Computer Networks

Network, Component and Categories, Topologies, Transmission Media, Reference Models: ISO/OSI Model and TCP/IP Model.

Unit II: Physical Layer

Digital and Analog Signals, Periodic Analog Signals, Transmission Impairments, Digital data transmission techniques, Analog data transmission techniques, Multiplexing and Spread Spectrum.

Unit III: Data Link Layer

Error Detection and Correction, Parity, LRC-CRC, Hamming Code, Flow Control and Error Control, Stop and wait, ARQ, Sliding window, HDLC, Multiple Access Protocols, IEEE 802.3 Ethernet.

Unit IV: Network Layer

Packet Switching and Datagram approach, IP addressing methods, Subnetting, Routing, Distance Vector Routing, Link State Routing, Broadcast and Multicast Routing.

Unit V: Transport and Application Layer

Transport Services, UDP, TCP, Congestion Control, Quality of Services (QOS), Application Layer: Domain Name Space (DNS), Electronic Mail.

Unit I: Introduction to Optical Fiber

Optical evolution, advantages of optical communication, its representations, Optical waveguides, basic optical laws, acceptance angle, numerical aperture Skew rays, Rays and modes, step-index, graded-index fibers, phase and group velocities, Signal degradation in optical fibers, attenuation units, absorption, scattering dispersions.

Unit II: Optical Sources

Optical source properties, finding the energy from the voltage, finding the frequency from the wavelength of light, operating wavelength of optical sources, semiconductor light-emitting diodes and laser diodes, semiconductor material and device operating principles, light-emitting diodes, surface-emitting LEDS, edge-emitting LEDS, super luminescent diodes, laser diodes, comparison of LED and ILD. Fiber optic transmitters, basic optical transmitters, direct versus external modulation, fiber optic transmitter applications, digital applications, analog applications.

Unit III: Optical Detectors

Light detectors, Role of an optical detector, Detector characteristics: Responsivity, Noise Equivalent Power, Detectivity, Quantum efficiency, Detector response time, Linearity, Spectral response, Noise considerations: Johnson noise, Shot noise, $1/f$ noise, Photon noise, The PN junction photo diode - PIN photodetector - Avalanche photo diode construction characteristics and properties, APD Specifications, Applications of APD - comparison of performance noise sources - simple model of photo receiver - Its equivalent for circulation of noise SNR, Optical Receivers.

Unit IV: Optical Amplifiers

Basic Concepts, Gain Spectrum and Bandwidth, Gain Saturation, Amplifier Noise, Amplifier Applications, Semiconductor Optical Amplifiers, Raman Amplifiers, Erbium-Doped Fiber

Amplifiers, System Applications - Optical Pre-amplification, Noise Accumulation in Long-Haul Systems, ASE-Induced Timing Jitter, Accumulated Dispersive and Nonlinear Effects, WDM-Related Impairments.

Unit V: Optical Switching & Networks

Transport Networks, Applications, Requirements, Architectures, Technologies, and Solutions, Introduction to Optical Access Networks, Optical TDM, subscriber multiplexing (SCM), WDM and Hybrid multiplexing methods.

Unit I: Introduction to AI

Control strategies, Search strategies, Production system characteristics, Specialized production system, Problem-solving methods, Problem graphs, Matching, Indexing, and Heuristic functions, Hill Climbing, Depth-first and Breadth-first, satisfaction, Related algorithms, Measure of performance and analysis of search algorithms.

Unit II: Knowledge Representation

Knowledge representation, Knowledge representation using Predicate logic, Introduction to predicate calculus, Resolution, Use of predicate calculus, Knowledge representation using other Logic-Structured representation of knowledge.

Unit III: Reasoning

Production-based system, Frame-based system. Inference: Backward chaining, forward chaining, Rule value approach, Fuzzy reasoning, Certainty factors, Bayesian Theory, Bayesian Network, Dempster-Shafer theory.

Unit IV: Game Playing and Planning

Overview, Min-Max search procedure, Alpha-beta cut-offs, Iterative Deepening, Components of planning system, goal stack planning, non-linear planning, hierarchal planning and other planning techniques, reactive systems.

Unit V: Understanding and NLP

Introduction to Understanding, Understanding as constraint satisfaction, Introduction to NLP, Syntactic and Semantic analysis, Statistical NLP, and Spell Checking.

Total Periods: 40